

Albin Babu Varghese

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EXPERIENCE

LensKit MAP Module | [lenskit/lkpy](#) | [#739](#)

- Designed and implemented the Mean Average Precision metric in `map.py`, adding 11 commits and integrating a new module under `lenskit.metrics.ranking`.
- Refactored rank computation to use `recs.ranks()`, added comprehensive test cases, and achieved 88% patch coverage.

Bun TextDecoder Fatal Option Fix | [oven-sh/bun](#) | [#19378](#)

- Fixed the `fatal` option in `TextDecoder` to correctly coerce 0, 1, `null`, and string values using `.toBoolean()`, resolving issue [#19362](#).
- Added and formatted tests in `test/js/web/encoding/text-decoder.test.js`, ensuring CI coverage and uniform test formatting.

Matplotlib Pie Error Fix | [matplotlib/matplotlib](#) | [#30019](#)

- Updated `axes.pie` to raise a `ValueError` when all wedge sizes are zero, preventing a NaN-to-integer conversion error as described in issue [#30007](#).
- Added a test in `test_axes.py` covering the all-zero case, ensuring the new error path is fully exercised.

VSCode-Python Extension Activation | [microsoft/vscode-python](#) | [#25025](#)

- Fixed issue [#25016](#) by adding `pylock.toml` to `activationEvents` in `package.json`, following PEP 751 naming conventions.
- Implemented a regex-based activation so the extension auto-activates on matching files.

Gitattributes Update | [rust-lang/rustc_codegen_gcc](#) | [#659](#)

- Added a `.gitattributes` file to treat `Cargo.lock` as text and normalize end-of-line characters, enabling GitHub to display diffs by default.

PROJECTS

TESV In-Game Coordinate Scraper | *Python, pymem* | [albus-droid/tesv-ingame-coordinates](#)

- Implemented a Python script to read real-time player coordinates (X, Y) and world state from Skyrim's memory.
- Output JSON-formatted telemetry at one-second intervals, including timestamp, coordinates, and Tamriel state.
- Managed memory offsets and process attachment via the `pymem` library for robust data extraction.

DApp Certification | *Ethereum, React.js, Solidity, Node.js, web3.js* | [albus-droid/dapp-certification](#)

- Developed a blockchain-based solution for educational certificate verification.

EDUCATION

Seneca Polytechnic, Toronto

PG Diploma in Cloud Architecture and Administration

Expected 2025

George Brown College, Toronto

PG Diploma in Applied AI Solutions Development

2024

GPA 3.39

APJ Abdul Kalam Technological University, India

Bachelor of Technology in Computer Science

2022

CGPA: 7.35/10 (9.5+ in Computer Labs)

Central Board of Secondary Education, India

Senior Secondary Education

2018

Aggregate: 88.6 (97/100 in Computer Science)

SKILLS

Languages: Python, Rust, JavaScript, Lisp, Bash, C, $\text{\LaTeX}$

Libraries/Frameworks: NumPy, Matplotlib

Tools: Linux, AWS, GCP, Azure, Git, GitHub Actions, Terraform, Ansible, Docker